Bihar Engineering University, Patna

B.Tech 4th Semester Examination, 2025

Course: B.Tech
Code: 104401

Subject: Analog Circuits

Time: 03 Hours
Full Marks: 70

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 ***Choose the correct option / answer the following (Any seven question only):*** **[2 x 7 = 14]**

(a) A cascode amplifier is a combination of
(i) CE–CE
(ii) CB–CB
(iii) CE–CB
(iv) CC–CE

(b) An amplifier has an input resistance of 2 kΩ. Using voltage-series feedback with $A\beta = 24$, the new input resistance is……..
(i) 2 kΩ
(ii) 25 kΩ
(iii) 50 kΩ
(iv) 62 kΩ

(c) Which MOSFET operates with zero gate voltage?
(i) Enhancement MOSFET
(ii) Depletion MOSFET
(iii) Power MOSFET
(iv) CMOS

(d) Hartley oscillator uses
(i) Two capacitors
(ii) Two inductors
(iii) One capacitor
(iv) One inductor

(e) If collector current increases from 1 mA to 4 mA, trans conductance increases by a factor
(i) 2
(ii) 4
(iii) $\sqrt{2}$
(iv) none

(f) Hybrid-π model is mainly used for
(i) DC analysis
(ii) Low-frequency analysis
(iii) High-frequency analysis
(iv) Power analysis

(g) The maximum efficiency of *a* class-B amplifier is
(i) 50%
(ii) 60%
(iii) 78.5%
(iv) 100%

(h) Two identical CE stages each have *a* bandwidth of 100 kHz. Overall bandwidth is approximately
(i) 50 kHz
(ii) 70.7 kHz
(iii) 100 6Hz
(iv) 200 kHz

(i) Which BJT configuration has the highest current gain?
(i) CB
(ii) CE
(iii) CC
(iv) Emitter follower

(j) Phase shift provided by the RC oscillator network is
(i) 90°
(ii) 180°
(iii) 270°
(iv) 360°

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Q.2 (a) A self-biased P-channel JFET has a pinch off voltage $(V_p) = 5V$ and $I_{DSS} = 12mA$. The supply voltage available is 12V. Determine the values of R_D and R_S so that $I_D = 5mA$ and $V_{DS} = 6V$. [7]

(b) Explain the frequency response of multistage amplifiers. [7]

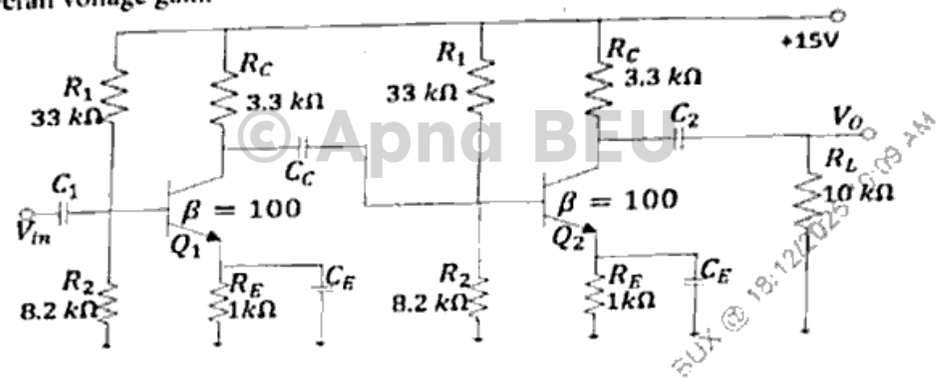
Q.3 (a) Explain the voltage-series and current-shunt feedback configuration. [7]

(b) Explain the circuit of the Hartley Oscillator in detail. [7]

Q.4 (a) Derive expressions for voltage gain, current gain, and output impedance of a CE amplifier using h-parameters. [7]

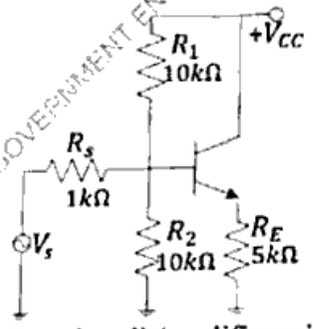
(b) Discuss various coupling schemes used in a multistage amplifier. [7]

Q.5 (a) For the RC coupled amplifier, determine the voltage gain of each stage and the overall voltage gain. [7]



(b) What is crossover distortion? How can it be eliminated? [7]

Q.6 (a) A transistor used in common collector mode has the following set of h-parameters: $h_{ic} = 2k\Omega$; $h_{fc} = -51$; $h_{rc} = 1$ *and* $h_{oc} = 25 \times 10^{-6} mhos$. Find the values of input and output resistances, current, and voltage gains of the amplifier. [7]



(b) Explain the working of a class-B push-pull Amplifier using a transformer, and prove that the efficiency of such an amplifier is ≤ 78.5%. [7]

Q.7 (a) Derive the small-signal model of the emitter follower and obtain expressions for gains and impedances. [7]

(b) Explain the limitations of multistage amplifier design. [7]

Q.8 (a) Explain the phenomenon of channel length modulation in JFET. [7]

(b) Explain the Miller effect in a CE amplifier using the hybrid-π model and derive the expressions for effective input and output capacitances. [7]

Q.9 Write short note on ***any two*** of the following: [7 x 2 = 14]

(a) Power amplifiers

(b) Colpitts oscillator

(c) Barkhausen criterion for oscillations

(d) Current shunt

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